

SPECIAL ISSUE PAPER

Effort estimation in agile software development: Case study and improvement framework

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Abstract

Effort estimation is more challenging in an agile context, as instead of exerting strict control over changes in requirements, dynamism is embraced. Current practice relies on expert judgment, where the accuracy of estimates is sensitive to the expertise of practitioners and prone to bias. To improve the effectiveness of the effort estimation process, the goal of this research is to investigate and understand the estimation process with respect to its accuracy in the context of agile software development from the perspective of agile development teams. Using case study research, 2 observations and eleven interviews were conducted with 3 agile development teams at SAP SE, a German multinational software corporation. The results reveal that factors such as the developer's knowledge, experience, and the complexity and impact of changes on the underlying system affect the magnitude as well as estimation accuracy. Furthermore, there is a need for a tool that incorporates expert knowledge, enables explicit consideration of cost drivers by experts and visualizes this information to improve the effectiveness of the effort estimation. On the basis of the findings of the case study, a framework, inspired by the quality improvement paradigm is proposed to improve effort estimation in agile development.

KEYWORDS

agile development, change impact analysis, effort estimation, framework, QIP

1 | INTRODUCTION

Problem statement

Effort estimation in Agile software development is challenging as the requirements are constantly evolving and are developed as the project progresses. As a result, the effort estimates need to be adjusted for every sprint to ensure delivery in time-boxed release. Each sprint involves planning, development, integration, testing, and delivery.¹ Incorporating a single change could entail a large impact across the various individual software artifacts, which makes it an important aspect to consider while planning and estimating effort. The systematic analysis of the impact of a change can be used to estimating its costs and facilitate planning.² Currently, methods based on subjective judgment are most frequently used in this context.^{3,4} In these methods, a team of experts typically estimates through consensus how much effort a certain change will entail. The accuracy of these expert-based methods is hampered by inconsistencies and wishful thinking.⁵ Because of the use of limited information and human judgment bias (individual and group effects), the prediction capability of this method is limited and error-prone.

Moreover, the current compositions of existing estimation methods (see Section 2) do not objectively consider the following, which makes the acquired estimates less reliable:

- The potential impact of a change on existing software artifacts.
- Factors that contribute to the effort overhead.

Therefore, improvement potential exists with respect to systematic effort estimation in this environment and marks the contribution of our research to the body of knowledge.

Most of the research on effort estimation in an agile context from real-world settings is based on a few surveys as described in Section 2. While this has helped to understand the broader state of the practice, there is value to be gained by conducting an in-depth analysis of this contemporary phenomenon in the agile context.

Objective

We intend to understand the current practice of effort estimation to improve the effectiveness of the effort estimation process in prevailing estimation methods with respect to estimation accuracy.

Contribution

Performing a case study with a case company in the,⁶ we investigated why and how estimation is done and what factors are considered relevant for making estimates and for improving estimation accuracy. The study further established the need for tool support and helped to identify the requirements for such a tool. This article is an extension of our paper⁶ and provides more details of the research methods used in the study. It also provides a framework that build on the results of the previous study. To improve effort estimation, the framework uses research on change impact analysis and binds it in the context of agile software development. It also exemplifies in detail the scenario where the suggested framework is operationalized.

Outline

The paper is structured as follows: Section 2 highlights the background and related work. Section 3 details the research study design with the performed case study. Section 4 presents the results, while Section 5 discusses and summarizes the major results of the case study where Section 6.1 introduces an early-stage design framework for improving effort estimation in agile development using change impact analysis. An instantiation of this framework is presented in Section 6.2. Section 7 concludes the paper while giving insights of future directions.

2 | RELATED WORK

A few secondary studies exist in the published literature that establish the state-of-the-art regarding different prospects of effort estimation in the context of agile development.

One secondary study⁴ reviewed the literature on estimation in agile, iterative, and incremental projects. It concluded that there is a lack of research on finding the impact of properties of historical/current project data on estimation results. Furthermore, it has established that more empirical validation of estimation models is required.

Another secondary study³ reviewed the literature on estimating effort in an agile context. The study concludes that the use of expert-based assessments is dominant and that there is lack of evidence on measuring prediction accuracy of proposed estimation techniques. It recommends considering factors other than size, for estimating effort.

A few surveys are also found in the published literature that investigated the state-of-the-practice regarding different perspectives of effort estimation in this context. A survey⁷ reported the results of interviews conducted with managers of 18 different companies and from 42 different projects in the Norwegian industry. The results showed that projects that adopted a flexible development process model experienced fewer effort overruns than the projects that adopted sequential development models. Using a dataset from a study by Moløkken-Østfold and Jørgensen,⁷ one study⁸ established that expert estimation is the dominant estimation method, that the use of formal models does not impact estimation accuracy and that managers are convinced of their high estimation accuracy, which in reality is not that high. The latter study did not, however, distinguish between projects using traditional or agile development models. Another survey⁹ conducted with 116 Chinese software organizations analyzing 112 projects found that large projects were more prone to cost and

schedule overruns and that 15% of organizations used model-based estimation methods with negligible benefit because of the high cost of adoption. This study also did not report whether the projects used traditional or agile development models. Another survey¹⁰ conducted with 60 agile practitioners from 16 different countries concluded that use of expert-based assessments is dominant, with effort being underestimated. It further reported that using a combination of estimation techniques may affect estimation accuracy positively, that estimation inaccuracy is mainly due to requirements and management related issues, and that team-related cost drivers are frequently used in agile development.

These studies have tried to establish the state-of-the-practice of effort estimation from different perspectives, but none of them has explored estimation of the impact of change requests with respect to their magnitude as well as the factors constituting effort overhead and affecting estimation accuracy.

3 | RESEARCH STUDY DESIGN AND EXECUTION

In this section, we have explained our research study design with respect to the objective of the study, the research questions investigated in Section 3.1. Further in Section 3.2, the design and execution of the case study is described where in Section 3.3 the validity threats are discussed.

3.1 | Objective and research questions

The objective of this study was to explore and understand the estimation process with respect to its accuracy in the context of agile software development from the perspective of agile development teams. To achieve this objective, we designed and performed an exploratory case study at a German multinational software corporation, SAP SE. A case study is useful for gaining a deeper understanding of a specific phenomenon in its natural setting.¹¹ Table 1 shows the research questions along with their objectives that were investigated and are briefly listed below:

RQ1:How is effort estimation currently performed (purpose, process)?

RQ2:What factors are considered relevant and important for improving estimation accuracy? The flexible design of the study allowed us to include RQ3 after performing the observation sessions, ie.

RQ3:For which aspects of the effort estimation process do practitioners require tool support?

It was observed that to know what is planned for a certain sprint, what about the availability and knowledge of a resource regarding a certain backlog item (BLI), whether there would there be any dependencies and so on, the teams had to hold discussions relying on the implicit knowledge they had in their minds. Additionally, they had to navigate and search manually for this information if it is stored in the tools they use in sprint planning. Because of these reasons, we included RQ3 to provide practitioners with some aspects of the estimation

TABLE 1 Research questions with objectives

Research Question (RQ)	Objective
RQ1: How is effort estimation currently performed (purpose, process)?	To understand the current estimation process used by the teams with respect to which estimation methods are being practiced—the purpose of estimation—the process detailing which tasks are performed, who participates, and what decisions are made on the basis of what criteria and rationale, as well as what information and tools are used. Also, to find the deviations if a standard method like Planning Poker is being used.
RQ2: What factors are considered relevant and important for improving estimation accuracy?	To investigate the current estimation process with respect to the following: • What factors are considered relevant for making estimates and to what extent? • What factors are considered important for improving estimation accuracy and to what extent?
RQ3: For which aspects of the effort estimation process do practitioners require tool support?	To investigate from the perspective of practitioners whether a certain aspect of the estimation process if supported by a tool would support them when making estimates and how.

TABLE 2 Unit of analysis

Characteristics	Team A	Team B	Team C
Team size	10	7	20
Team experience with software development (in years)	14	18	10
Team experience with agile development (in years)	4	4	5
Agile methods being used	Scrum, TDD, XP	Scrum, TDD	Scrum, TDD, XP
Programming language	ABAP	Javascript	Javascript

Abbreviations: TDD, test-driven development; XP, extreme programming.

process that, if they are tool-supported, may support them when making estimates. The research questions with their objectives can be found in Table 1.

This case study was performed and documented according to the guidelines of Runeson and Höst.¹¹ The outcomes of this study will be used directly as a basis for developing a methodology and a tool for estimating the effort of implementing changes in the context of agile software development as well as for evaluating the methodology and the tool in an industrial setting.

3.2 | Case study design

We chose a multiple-case design to get more in-depth insights on how estimation has been performed in agile software development. A multiple-case study is a variant that investigates at least 2 cases of the same phenomenon. Thus, it is especially useful for investigating a phenomenon across different settings and for exploring how the results vary among them.¹²

3.2.1 | Case study context

The case study was performed with a German multinational software corporation, SAP SE (Systems, Applications & Products in Data Processing). SAP SE develops enterprise applications in terms of software and software-related services to manage business operations and customer relations. Headquartered in Walldorf, Germany, SAP has its regional offices in 130 countries. On the basis of market capitalization, SAP is the world's third largest independent software manufacturer. The company has over 250,000 customers in 190 countries.¹³

To keep pace with the changing market trends and customer needs, the company has also integrated agile practices such as XP¹⁴ and

Scrum.¹⁵ Because we were interested in exploring the estimation process in an agile context, we therefore focused on ongoing projects practicing agile principles.

3.2.2 | Case and unit of analysis

Studies on effort estimation show that Scrum and XP are found to be the most frequently-used agile methods for making estimates on the sprint or release planning level.^{3,4} Therefore, we narrowed the context to the Scrum and XP methodology and defined the case to be studied as "Scrum and XP teams performing estimation while planning for a sprint". We intentionally selected the unit of analysis to allow the comparison of deviant cases such as estimation method (individual vs group estimation) and team size (small vs. large). Accordingly, we drew a sample of 3 units of analysis corresponding to 3 different teams. Table 2 shows the units of analysis. The experience of the teams is aggregated and is shown as median value. The teams description is provided in Table 3. As observed from Table 3, Scrum, test-driven development and extreme programming (XP) are being practiced by all 3 teams. All teams are collocated, delivering 12 sprints per year, with a sprint lasting for 4 weeks.

The case study protocol used in this research includes (1) the procedures for selecting the units of analysis and making arrangements for the field work; (2) the informed consent for protecting the participants and the organization; (3) the data collection procedures, forms, and plan; and (4) the data analysis procedures.

3.2.3 | Data collection

The case study was performed during the second half of 2014. To reduce bias and increase the credibility of the empirical data and

TABLE 3 Teams

Team	Description
Team A	It is comprised of 10 members with an average experience of 4y of agile development and 14y of software development and is associated with the development or enhancement of a web application user interface (UI) framework based on Web Dynpro ABAP technology. Web Dynpro is the SAP standard UI technology for developing Web applications in the ABAP environment. ABAP (Advanced Business Application Programming) is a high-level programming language developed by SAP.
Team B	It is comprised of 7 members with an average experience of 4y of agile development and 18y of software development. It is associated with the development or enhancement of a UI control, ie, SAP Visual Business. SAP Visual Business visualizes and combines enterprise data with geographical or 3D real-world scenes. Its relation to reality allows users to navigate intuitively through business data.
Team C	It is comprised of 20 members with an average experience of 5y of agile development 10y of software development. It is associated with the development or enhancement of a Web application UI framework on the basis of the SAPUI5 toolkit. SAPUI5 development toolkit is a UI framework built on top of JQuery and open Web Standards like HTML5, CSS3, and LESS. It provides SAP's new enterprise-ready HTML5 rendering library for client-side UI rendering and programming.

findings, multiple data sources were used following a triangulation approach.

The first source driving the qualitative analysis was observations sessions of sprint planning meetings to watch the effort estimation process. The second source was a set of interviews. Data collection was performed as follows:

Introduction meeting: A meeting was held to introduce the purpose, process, and outcomes of the case study to the Scrum masters and product owners (PO) of the 3 selected teams. The purpose of the meeting was to enable the teams to make an informed decision about accepting or declining their participation in the study. All questions and concerns were resolved. After signing the informed consent, the participants and the researchers jointly schedule the observation and interview sessions. The informed consent included the research objective, study plan, data collection procedures, storage and use, access to raw data and results, planned publications, and procedures for anonymity and confidentiality. Scrum masters and POs acted as representatives of their teams and were responsible for disseminating this information within their teams.

Observations: Two researchers observed the sprint planning meetings of the teams. They independently recorded data regarding the participants (roles and responsibilities), their interactions, the estimation process (inputs, steps, rationale, and outputs), and the tool used. After each observation, they compared their notes and generated a consensual observation protocol.

Interviews: Interviewees were selected so that the main roles in the team were covered, ie, PO, Scrum masters, and at least 1 developer be interviewed. However, the number of interviewees varied according to the team size. In Teams A and B, there were 3 and in Team C, there were 5 interviewees.

The design of the interview consisted of 3 main parts, and the duration of the interviews was 45 minutes to 1 hour. In the first part of the interview, an introduction of the study goals was provided. Furthermore, the interviewees were informed why they had been selected for the interview. It was also ensured that all the information they provided would be treated confidentially. In the second part, the interviewees were asked about their experience in general and their experience with the agile development approaches in particular. They were also asked about their role and responsibilities at SAP and their working

domain, platform, and language, and about effort estimation concepts in general. In the third part, questions related to effort estimation in particular were asked in a semistructured interview.

Altogether, 11 interviews were performed either face-to-face or via video conference. Two researchers were involved in conducting the interviews: One performed the interview and the other took notes. All interviews were recorded and transcribed. After each interview, they compared their notes with the electronic record, and generated a consensual interview protocol. The interview questionnaire was derived by operationalizing the research questions using the goal-question-metric approach.¹⁶

3.2.4 | Data analysis

The data was analyzed as follows:

Observations and interviews: We performed within-case and cross-case analyses. We first analyzed the data within each case (ie, in each team). Then, we compared, related, and integrated the results across the other cases (ie, across teams). Performing these analyses gave us opportunity to compare deviant cases with respect to estimation method (individual vs group estimation) and team size (small vs large). Qualitative data analysis was performed by using coding techniques and MAXQDA.¹⁷ Coding here refers to the process of identifying and recording 1 or more text passages that can be linked to a code (ie, a well-defined concept such as a specific task, process, role, or interaction). We coded observations and interviews using a predefined list of codes (derived from the research questions and literature). If necessary, we extended the list of codes during the analysis. Quantitative data analysis were performed using IBM SPSS Statistics.¹⁸ We report descriptive statistics including the sample median (Mdn), minimum (Min), maximum (Max), frequencies, and valid percentages. For 5-point rating scale data, we were interested in which items produced a meaningful opinion from the respondents, ie, a response significantly different from neutral. To do this, we used the 1-sample Wilcoxon signed rank test to test for significant differences from the midpoint ($H_0: \text{Mdn}(x) = 3$).¹⁹ In these cases, we report the 1-sample Wilcoxon signed-rank observed value (Z) and the significance level (p). The raw data and analysis results are stored in Fraunhofer IESE repository. The raw data cannot be shared because of the confidentiality agreement.

Feedback: We aggregated, presented, and discussed the results with the corresponding teams. This approach was used to increase the trustworthiness of the results and ensure that they reflect the participants' insight. The participants reviewed the results and agreed with the findings.

3.3 | Threats to validity

We evaluated the case study using the checklist proposed by Runeson and Höst¹¹ to strengthen the validity of the results. This ascertained that we addressed all the requirements of case study design and execution including study objective, design protocol, unit of analysis, data collection, and analysis methods, among others. In the following, we will discuss some threats and the ways we tried to mitigate them.

Construct validity: To strengthen the empirical evidence, multiple data sources (interviews and observations) were used following a triangulation approach.

To mitigate the risk of inappropriate measurement, we used the goal-question-metric approach and performed peer reviews with another expert in empiricism to systematically operationalize variables and design data collection instruments. We asked the expert to review the understandability, completeness, and validity of the data collection procedures and instruments and give feedback. This feedback also enhanced the quality of the case study design. Additionally, we avoided the risk of mono-operation bias by designing a multiple-case study. We focused on deviant cases regarding the estimation method (individual vs group estimation) and team size (small vs large). This allowed us to investigate extreme cases.

Internal validity: To avoid bias in study results, only those teams were approached that had sufficient experience in agile methodologies. During the interviews, the practitioners were asked about their experience specifically in agile development in addition to their experience in software development in general.

To avoid diffusion bias, we instructed participants not to share interview experiences or data with other members of their team or across teams. They also signed an informed consent. After analyzing the data and the observation and interview protocols, we did not find any indication if participants had shared their experiences or results.

Furthermore, to avoid evaluation apprehension, the practitioners were assured of the anonymity of their personal and organizational data. We also faced slight deviations from the data collection protocols because of time constraints and low staff turnover. For example in Team A, we could not interview their PO and could not observe their Scrum planning and estimating session. We mitigated this threat by conducting the feedback session where the teams reviewed the study results and agreed with the findings. In case studies, researchers may (un)intentionally make subtle comments that affect the participants behavior or answers. We avoided this risk by using standardized data collection methods and instruments. During the interviews, 1 researcher acted as an observer to monitor and document any influence of the interviewer. After analyzing the observation and interview protocols, we did not identify any experimenter bias.

External validity: As our case study used a convenient sample, our results might not be generalizable. However, they may be repeatable in a similar context and with similar characteristics with respect to agile development. To compensate for the small sample, we intend to design and execute a survey to investigate effort estimation processes in the German software industry employing agile development processes.

Conclusion validity: To ensure reliability in the study execution, all steps of the study were performed by the same researchers, following the same procedures, using the same instruments. Moreover, researchers maintained detailed memos of the study execution and any possible confounding factors. Two researchers performed all the quantitative and qualitative analyses independently and all the discrepancies were discussed and resolved, if found any. These measures helped to reduce bias and to increase the credibility of the results.

We aggregated the results of the teams and only used median values, and the number of respondents to identify common practices or views and to point out potential future research areas. However, it is important to remark that further statistical analysis were not possible because of the small sample size. Thus, further such studies are needed to allow statistical analysis and the generalization of results.

4 | RESULTS

In Section 4.1, we present the results regarding the estimation method being followed. Then in Section 4.2, factors affecting estimation accuracy are detailed where Section 4.3 highlights the aspects of the process that require tool support.

4.1 | Estimation method (purpose and process)

RQ1: How is effort estimation currently performed (purpose and process)?

All the teams do estimation in Scrum sessions at the sprint planning level. The estimates are made for requirements at the granularity level of BLIs, which is a fine-grained level of a user story. The teams use MS Excel and the issue tracking and project management tool, JIRA.²⁰ The teams do estimation for scheduling tasks but not for risk assessment,²¹ project bidding, or budgeting²² for example.

Other than that, the teams perform estimation to develop a common understanding among them regarding what needs to be done in a sprint. Teams A and B do individual estimations, estimating "Effort in Person hours." Team C plays the estimation game²³ where estimates are provided through consensus. Team C estimates "Complexity of BLI in Story points."

4.2 | Estimation accuracy factors

RQ2: What factors are considered relevant and important for improving estimation accuracy?

To investigate what factors are considered during estimation, we captured the interviewees' perspective in 3 different ways. First, the interviewees were asked in an open-ended question to provide factors they consider (implicitly in their minds) while making estimates. Figure 1 shows the factors with the number of practitioners providing them. Developer implementation experience was considered most

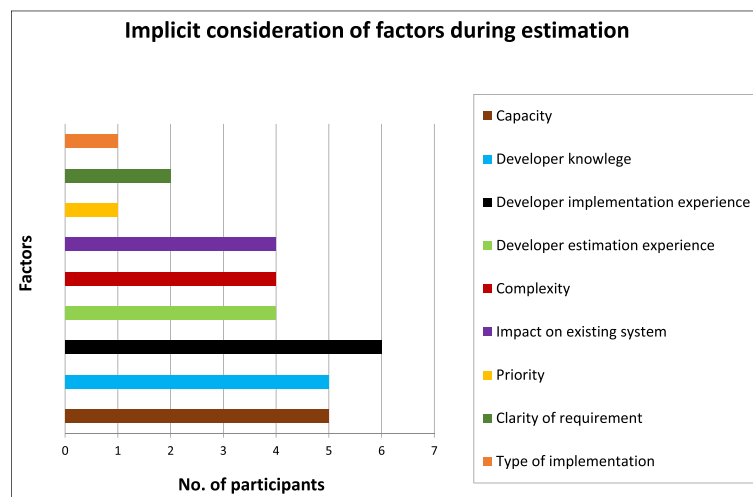


FIGURE 1 Factors considered implicitly by practitioners while making estimates

TABLE 4 Factor definition

Factor name	Factor definition
Developer's experience with making estimates.	Refers to the experience of a developer with estimating development tasks.
Developer's knowledge of BLI.	Refers to the knowledge of a developer regarding the system or component he or she is working on as well as a certain BLI to be implemented in that system or component.
Developer's experience of implementation.	Refers to the past implementation experience and familiarity with estimated system or component of a developer as well as the current system or component he or she is working on.
Dependencies among BLI.	Refers to existing dependencies (coupling with other BLIs) among the BLIs to be implemented in a sprint.
BLI complexity.	Refers to the complexity of a new BLI in terms of its type, ie, functional or nonfunctional, number of required inputs or outputs, internal or external interfaces, design constraints, user and location, and required feature. ²⁴
BLI impact on parts of the developed system.	Refers to the impact of implementing a new BLI on the existing system, ie, identification of system classes or components affected by the change or the number of test cases required, for retesting the system after a change etc.
Estimates made for a similar BLI in previously completed sprints.	It means a similar (in terms of actual and estimated effort) BLI was already estimated and implemented in the past, so both its (1) actual effort and (2) estimation error should be known.

Abbreviation: BLI, backlog item.

relevant (provided by 6 out of 11 practitioners) followed by Developer knowledge and Capacity. According to the teams, *Capacity* means the amount of work load a certain developer has. Furthermore, "Clarity of BLI" and "Type of implementation" were mentioned once by 2 different interviewees, respectively.

Second, the interviewees were presented with 7 factors (see Table 4) to rate the extent to which each factor is relevant for the estimation process (scale: 5-point ratio scale, with 5 being very relevant and 1 not being relevant at all). There was also an opportunity to provide additional factors they thought were relevant and rate them accordingly. However, no one provided any additional factor. Table 5 shows the aggregated results of the teams (ie, the median values of all the interviewees in the middle column) rating each factor with respect to its relevance for estimation. Almost all factors are rated as relevant to be considered for the estimation process. However, the factor

Estimates made for a similar BLI in previously completed sprints does not seem relevant for making estimates.

Finally, the interviewees were presented with the same 7 factors (see Table 4) to rate the extent to which each factor has the potential to improve estimation accuracy (scale: 5-point ratio scale, with 5 being very relevant and 1 not being relevant at all). There was also an opportunity to provide additional factors they thought could improve accuracy and rate them accordingly. However, no one provided any additional factor. Table 6 shows the aggregated results of the teams (ie, the median values of all the interviewees in the middle column) rating each factor with respect to its potential to improve estimation accuracy. Here again, in Table 6, almost all factors were rated as having a high potential to improve estimation accuracy. However, the teams apparently were unsure whether having estimated similar BLI in the past (ie, the factor *Estimates made for a similar BLI in previously completed sprints*) has any effect on improving estimation accuracy.

TABLE 5 Factors considered relevant for estimation

Factor name	Minimum value	Median value	Maximum value
Developer's experience with making estimates	1	4	5
Developer's knowledge of BLI	1	5	5
Developer's experience of implementation	1	5	5
Dependencies among BLI	2	4	5
BLI complexity	3	5	5
BLI impact on parts of the developed system	3	4	5
Estimates made for a similar BLI in previously completed sprints	2	2	4

Abbreviations: BLI, backlog item.

TABLE 6 Factors considered important to improve estimation accuracy

Factor name	Minimum value	Median value	Maximum value
Developer's experience with making estimates	2	4	5
Developer's knowledge of BLI	4	5	5
Developer's experience of implementation	3	4	5
Dependencies among BLI	2	4	5
BLI complexity	0	4	5
BLI impact on parts of the developed system	2	4	5
Estimates made for a similar BLI in previously completed sprints	2	3	5

TABLE 7 Tool Aspects

Tool Aspect ID	Aspect name	Aspect definition
TA 1	Influential factors	Includes providing information for factors like in Table 4, eg, "developer's experience" that can influence the estimation accuracy.
TA 2	Impact analysis	Includes analyzing the impact of implementing BLI on the existing system. This information can help identify the number of system classes or components affected by the change or the number of test cases required, for retesting the system after change.
TA 3	Performing estimation	Includes providing development teams with an initial estimate, including the range of estimation uncertainty. To support team in deciding on final estimates, the method and tool will support analysis of influences and impacts as specified in TA 1 and TA 2.
TA 4	Visualizing estimation	Includes visualization of the aspects TA 2 as well as TA 3. This will aid in identifying impacted components by implementing a certain BLI as well as performing risk analysis if certain BLI cannot be finished in a time-boxed sprint.
TA 5	Expert interaction and updating estimation	Includes a support for an interaction between estimation support and experts, in which estimation expert adjusts the information initially provided by the tool (eg, identified impacts) and the tool provide revised analysis, estimates, and visualizations in response.

Abbreviations: BLI, backlog item; TA, tool aspect.

4.3 | Tool support

RQ3: For which aspects of the effort estimation process practitioners require tool support?

The interviewees were asked to assume that they were interacting with a tool for estimating effort. They were presented with 5 aspects (see Table 7) of the effort estimation process where they might require tool support and could improve the estimation process as a whole. They were asked to choose those aspects that they thought require tool support and also explain the why a certain aspect might improve the estimation process. The aspects are explained below in Table 7.

Here, it is important to mention that out of all factors as listed in Table 4, the factor *BLI impact on parts of the developed system* is treated as a separate aspect called *Impact analysis*, which requires tool support. According to the definition in Table 4, we think that

the identified system classes/ components potentially impacted by a change would prove more useful for the estimation process if they are visualized, allowing the expert to interact with them. This would provide more transparency and awareness and would support the expert in making informed decisions for estimating the impact of a certain change. All the other factors are grouped under the aspect *influential factors*.

The explanations the participants provided on how a certain aspect could improve the estimation process were categorized as follows:

1. Required: a certain aspect requires tool support.
2. Required with conditions: a certain aspect requires tool support under given conditions.
3. Counterproductive: a certain aspect, if supported through a tool, could be counterproductive.

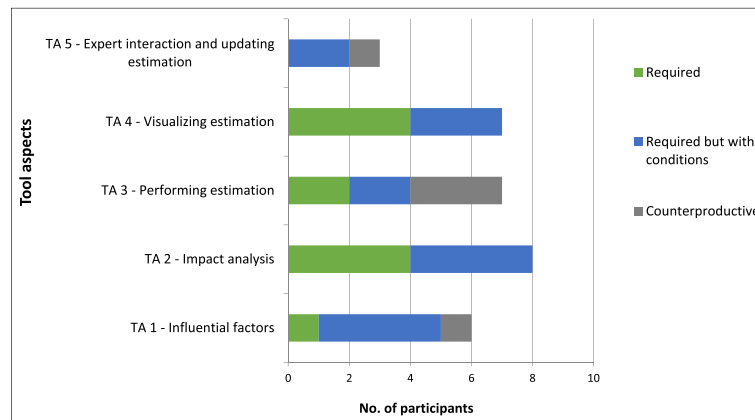


FIGURE 2 Aspects that require tool support

Some insights of the interviewees regarding some of the tool aspects (TA) are as presented below:

TA 1: “Influential factor”: 5 out of 11 participants chose it as either *Required* or *Required with conditions*. According to these participants, “...TA 1 provides awareness of influential/ non-influential factors, ...TA 1 could have a positive impact on accuracy if historical data is available and if it is done by whole team ...”

TA 2: “Impact analysis”: 8 out of 11 participants chose it as either *Required* or *Required with conditions*. According to these participants, “...TA 2 provides awareness of dependencies (e.g. prerequisites, contact persons, team involved, start/end day), ... allows parallel implementation of independent BLI ... supports better understanding of complexity...”. Moreover, “TA 2 may improve estimation process provided if data is up to date and dependencies exist among BLIs...”

TA 3: “Performing estimation”: 4 out of 11 participants chose it as either *Required* or *Required with conditions*. According to them, “...TA 3 provides training of developers in making estimation... supports PO for prioritizing BLIs... also it may improve the estimation process provided that it “...serves as starting point for the estimation... provides awareness of individual estimation performance... supported by visualization...”.

TA 4: “Visualizing estimation”: 7 out of 11 participants chose it as either *Required* or *Required with conditions*. According to them,

“...TA 4 supports risk management ... supports planning of estimates...”. Moreover, it may improve estimation process provided that “...data is up to date... provides visualization of team capacity and the estimation of BLIs together...”.

On the other hand, for **TA 1:** “Influential factors,” participants who chose this aspect as *Counterproductive* described the possible reasons for their opinion ie, “...estimates provided by the tool may become risky if one person is rating TA 1 alone...”

and for **TA 3:** “...TA 3 could keep them away from thinking and discussing pro-actively in the team...”.

Figure 2 shows the aspects that required tool support and the number of participants choosing each aspect as either *Required*, *Required with conditions*, or *Counterproductive*.

The interviewees were also asked to provide additional aspects they thought require tool support and would improve the estimation

process. Two out of 11 interviewees came up with requirements such as “Tool should be simple to use” and “Proper integration of tool and data” instead.

5 | DISCUSSION

In Section 5.1 we have discussed the estimation method being followed by the case company. Then in Section 5.2, factors affecting estimation accuracy are discussed where Section 5.3 the aspects of the process that require tool support are discussed.

5.1 | Estimation purpose and process

The teams have a clear process for organizing estimation sessions and they actively participate in them. Moreover, it seems that developing a common understanding among the team members is more important than the sole purpose of planning effort or schedule. It was also observed that the estimates made and the actual effort spent are only available till the end of a sprint. These estimates are revisited and reworked at times for the assignment of incomplete BLIs to the next sprint. They are not used for the purpose of measuring effort estimation accuracy. One reason for not storing this information is that the practitioners were satisfied with the accuracy of their current estimates. However, this hinders objective reflection on the practice of effort estimation in the teams. Without this awareness, the current estimation may affect several of the tasks such as project planning and scheduling as well as the development of a common understanding in teams.

Since the impact of estimation performance on project performance (eg, schedule slippage) is unknown, this may prevent PO from making commitments to external stakeholders regarding BLI delivery in a planned sprint.

Storing estimates will help the teams to track, learn, and reflect on estimation performance during the retrospective meeting and eventually improve the effectiveness of the estimation process. With this awareness, the experts would take informed decisions and could make commitments to external stakeholders. Finding this to be an improvement opportunity, we discussed it with the teams during the feedback session. The participant’s agreed with the suggestions and gave positive feedback.

TABLE 8 Factors relevance for estimation method versus number of practitioners considering them in current estimation method

Factor name	How relevant are the factors considered for the estimation method (median value)	Number of participants implicitly considering the factor while making estimates
Developer's experience with making estimates	4	4
Developer's knowledge of BLI	5	5
Developer's experience of implementation	5	6
Dependencies among BLI	4	0
BLI complexity	5	4
BLI impact on parts of the developed system	4	4
Estimates made for a similar BLI in previously completed sprints	2	0

Abbreviations: BLI, backlog item.

5.2 | Estimation accuracy factors

During analysis of the estimation factors, we came across interesting findings. We compared the factors rated as relevant for estimation by all 11 participants with the number of practitioners considering them implicitly in the current estimation method. Table 8 shows this comparison. It was found that factors that are rated as highly relevant for making estimates are either considered to a limited extent or are not considered at all during the estimation process. For example, in Table 8, the factor "BLI impact on parts of the developed system" is rated as relevant where as only 4 out of 11 practitioners are implicitly considering it. Similar cases are with the factors "Developer's experience with making estimates" and "BLI complexity" for example. These factors are also rated high for improving estimation accuracy but are being implicitly considered to a limited extent (see Table 6).

On the other hand, the factor Dependencies among BLI is rated as relevant, where as no one is implicitly considering it in Table 8. Furthermore, it can be noted from Table 6 that it is also rated as high for improving estimation accuracy. This shows that when provided with a factor explicitly, the practitioners realized this factor's relevance and impact on accuracy. This explicit consideration of factors is missing in the currently-used estimation methods.

Interestingly, the factor Estimates made for a similar BLI in previously completed sprints is neither rated as relevant (median = 2) nor any participant implicitly considers it (see Table 8). One possible reason for this could be the practice of not storing estimates as a result no one realizes the impact it can have on estimation accuracy. This can also be observed in Table 6, which shows that the practitioners were unsure of the impact of this factor on effort estimation accuracy.

We also collected data from the perspective of the individual teams to see if there are any deviations from the aggregated results, but did not find any. The factors were considered important among all the teams alike independent of their size, experience, or practiced estimation method.

The explicit provision and consideration of effort factors currently missing in the estimation methods were identified as an improvement opportunity. We discussed this too with the teams in the feedback session. Participants agreed that considering the factors including Dependencies among BLI, Estimates made for a similar BLI in previously completed sprints and BLI impact on parts of the developed system among others can improve the effectiveness of estimation process.

5.3 | Tool support

It is observed that currently estimation is based on partial information due to the unavailability of information necessary for BLI estimation, for example, BLI impact on existing system because current tools (JIRA, MS Excel) do not provide it and are incapable of generating it. It was also interesting to see that practitioners agree on certain aspects for which they need tool support and by going 1 step forward also provided insights in Section 4.3 on how and under what conditions these aspects may or may not help improve the estimation process overall.

Since the factor BLI impact on parts of the developed system was observed as being partially considered and was also rated as relevant for improving estimation accuracy, visualization of this factor as an interactive aspect would improve the effectiveness of the estimation process. And thus, most practitioners agreed that the tool support for the aspect Impact analysis is necessary and would prove very useful.

On the other hand, few participants chose *Performing Estimation* aspect as being counterproductive, saying that it will keep them from discussing proactively in the team and coming up with a common estimate and understanding. This could be because developing a common understanding in team was considered more important than making estimates. Using this aspect, the participants might lose the opportunity to discuss issues among themselves. To clarify this misinterpretation, we provided the intended purpose of this aspect in the feedback session. The purpose of this aspect was to provide an initial estimate with a margin of error to give team a basis for their discussion regarding what needs to be taken care of before coming up with a final estimate. This aspect aims to incorporate the other aspects that are either missing from the current estimation process, such as *Impact analysis* or that are not considered explicitly such as *Influential factors*. The practitioners responded with positive feedback regarding the aspects of the tool.

5.4 | Summary of the case study

The major results include factors such as developer's knowledge, experience, and the complexity and impact of changes on the underlying system affect the magnitude as well as the accuracy of estimation. Moreover, if data on estimates is stored, it can help to improve the estimation performance. Furthermore, the results indicate that certain

What: Analyze and operationalize IA using different development artefacts to improve effort estimation

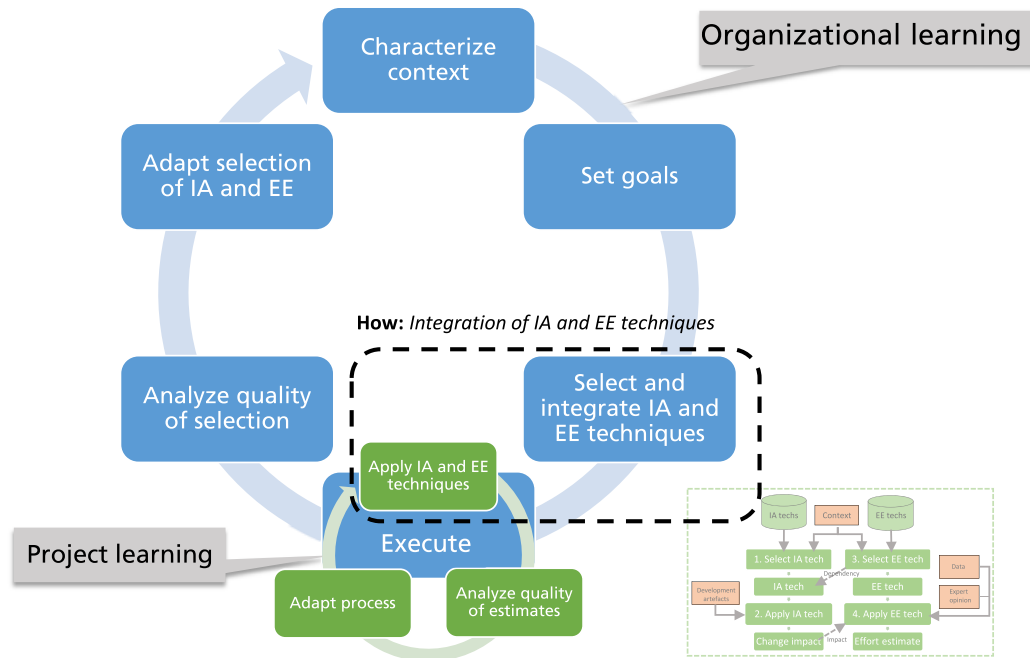


FIGURE 3 Framework for integrating impact analysis (IA) with effort estimation (EE) and improving effort estimation

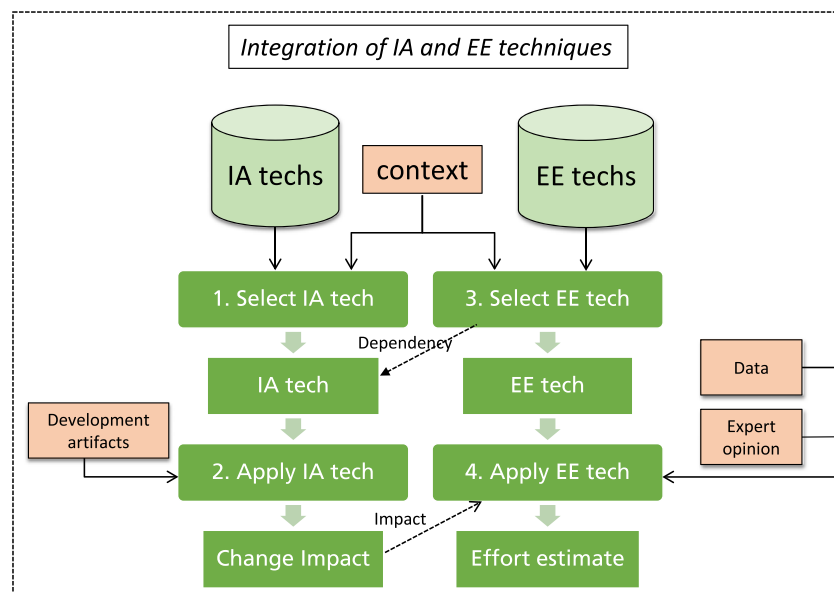


FIGURE 4 Integration of Impact Analysis (IA) with Effort Estimation (EE) techniques

aspects of the estimation process, such as the potential impact of a change on the underlying system, can improve estimation effectiveness if they are supported by a tool.

None of the existing estimation methods (in traditional or agile development) have so far considered quantifying the impact that a change has on existing software artifacts. Furthermore, to collect and analyze these effort factors a huge amount of data and cost are required in data-driven methods like COCOMO I.²⁵ In hybrid methods like CoBRA[®],²² however, these effort factors are considered, but they

need to be adapted to the agile context. We therefore aim to develop a methodological framework that explicitly considers these factors and as well as quantifies the impact of changes for estimating the effort required for changes more effectively in the agile software development context. We describe our framework in the next Section 6.

6 | IMPROVEMENT FRAMEWORK

In Section 6.1, we introduce our improvement framework along with its hypothetical application scenario in Section 6.2. Although hypothetical, the scenario is based on active discussion with an expert in effort estimation at Fraunhofer IESE and an understanding of the current work of

* CoBRA[®] is a registered trademark of the Fraunhofer-Gesellschaft and stands for cost estimation, benchmarking, and risk assessment.

the estimation process in the case company, it serves as an illustrative example to demonstrate the operationalization of the framework. This is 1 of the scenarios that we plan to evaluate with the case company along with the framework.

In our case study, a concrete need for supporting effort estimation methods with change impact analysis data was identified. Moreover, because of the incremental and iterative nature of agile software development, it can be greatly beneficial to consider the impact of new requirements or changes on existing artifacts and activities. Change impact analysis is a technique that identifies the effects of a change or estimates the tasks required to implement a change.² It can be used for understanding a program, predicting the impact of a change, and estimating the costs of change before implementing it.² However, existing effort estimation techniques used in the context of agile software development do not leverage the tools and techniques that can perform change impact analysis and that make it easier for practitioners to make accurate effort estimates. Therefore, to leverage the support provided by change impact analysis techniques and tools, we identified the need for a framework to guide the use of change impact analysis for improved effort estimation.

6.1 | Framework

The framework proposed in this paper supports researchers in selecting appropriate change impact analysis and effort estimation techniques in a given organizational context. Additionally, it supports practitioners in improving their estimates by using and integrating change impact analysis in their estimation process. By reusing knowledge and experience gathered in each iteration, the framework, which is inspired by the quality improvement paradigm (QIP),²⁶ enables continuous improvement towards effective effort estimation in practice. The overall framework and process is depicted in Figure 3 and Figure 4. The 6 steps of QIP applied in our context (agile development) and in the approach are as follows:

1. **Characterize context:** In this step, the organizational and project context with respect to effort estimation and impact analysis technique is specified. For example, the practitioners specify the estimation purpose, level of planning, prevailing estimation method, personal factors (experience, knowledge), estimation trend, estimation error, available development artifacts, and their level of granularity (eg, requirements or user story or BLI), type of change, tool support, requirements on tool, etc. This information is stored in an experience-base to be reused later on in sprints.
2. **Set goals:** In this step, the practitioners specify a measurable or quantifiable or concrete improvement goal for the estimation process. For example, the goal may be to reduce the error margin in effort estimates by 20%.
3. **Select impact analysis (IA) and effort estimation (EE) techniques:** The process of selection, application, and integration of IA and EE techniques is shown in Figure 4. In step 1 and step 3 of Figure 4 taken from a collection of IA and EE techniques, the practitioners can select appropriate IA and EE techniques, respectively, on the basis of their context, using our guidelines. It is important to note that the process of selecting IA and EE techniques can be either sequential or parallel. This is the case because not only the selection of both EE and IA is driven by the context, but the selection of EE can also be dependent on a certain selected IA technique. In the

latter case, the process of selection will be sequential and not parallel. We are investigating this dependency relationship to explore these possibilities and constraints. To make this integration possible, we first established the feasibility of using impact analysis while doing effort estimation in the agile context. Next, we need to capture the requirements for integration including finding the dependencies of IA techniques on EE, and finally provide some guidelines on how to perform this integration.

4. **Execute:** The process of applying and integrating IA and EE techniques is executed as follows:

- (a) **Apply and integrate IA and EE techniques:** In this step, first, the selected IA technique is applied on the given artifacts to find the impact and then it is integrated with EE technique, using this impact as input to EE, to find the estimates. The integration of IA with EE techniques is facilitated through a set of guidelines that we are working on. The initial estimate produced as a result of this integration is further refined reusing previous knowledge and data gathered during the sprints. As mentioned earlier in 3, this selection of EE can be dependent on the selected IA technique. For example, if from a selected IA, a practitioner is able to quantify the size and complexity of a change, she can select data-driven, hybrid techniques instead of pure expert-based techniques for making estimates on the basis of this information. However, if she cannot quantify the impact of a change but can still assess the software parts impacted by the change, then she can only select pure expert-based estimation methods based on this information to make estimates. We further need to investigate whether impact size and complexity could be quantified using code metrics such as lines of code or cyclomatic complexity (eg, using McCabe), respectively, or whether these could be rated by the practitioner on the basis of her experience. In either case, the impact information serves as input to the EE technique to produce estimates. The information regarding selected techniques and produced estimates is also stored in the experience base.
- (b) **Analyze quality of estimates:** The estimates made are assessed by the practitioners at the end of a sprint in the retrospective meeting or estimation workshop. The estimated values are extracted from the experience base and compared to the actual ones. In case of deviations, further investigation is required to find the cause of the deviation and feedback is provided to minimize the estimation error. To do this analysis, other data such as historical effort data, data regarding change requests, impact data, and effort overhead (factors) is required in addition to expert opinion and is retrieved from the experience base.
- (c) **Adapt process:** In this step, the feedback gathered in a study by Nguyen-Cong and Tran-Cao 4 is considered to adapt the process for the next sprints. To adapt the integration process, effort factors (as also identified by our case study), previous knowledge of the practitioner, and the data gathered in Nguyen-Cong D, Tran-Cao 4 are considered. In addition, any further requirements regarding the guidelines for integration

are also elicited from the practitioners. Using the parameterization of the estimation process based on factors and the feedback of the expert, the estimation process is adapted for the next sprint. This knowledge is also stored in the experience base and is reused in the next sprint. Through this small circle, practitioners learn about their project and estimation performance over sprints.

5. **Analyze quality of selected techniques:** At the end of a project release, the selected IA and EE techniques are analyzed regarding their suitability and effectiveness for the estimation process and whether they could achieve the specified goals. Information like feedback of the practitioners, stored estimation data, and knowledge, as well as the adaptations to the integration process over sprints is required and is retrieved from the experience base to analyze the effectiveness of the selected IA and EE techniques.
6. **Adapt selected techniques:** In this step, if the selected techniques are found unsuitable or ineffective and the set goals are not achieved, practitioners discuss and explore other techniques from the collection of IA and EE techniques that might be suited better in a given context. Through this cycle, the organization learns about their projects over time. In our agile development context, organizational level learning refers to experience and knowledge gathered from release to release, whereas project-level learning refers to knowledge acquired from sprint to sprint. The feedback can be provided in the context of retrospective meetings or by arranging additional estimation workshops—whichever is feasible for the organization.

The Steps 3 and 4 are the major building blocks of our framework. On the basis of the context, the practitioners can select and integrate IA and EE techniques using our guidelines in these steps. The practitioners can then use impact analysis while making estimates in order to improve them. The next section describes an example application scenario of this framework.

6.2 | Framework instantiation (application scenario)

In this section, we will consider a hypothetical application scenario, an instantiation of our framework.

1. **Characterize context:** The context is as follows:

- *Available artifacts:* Requirements (at the granularity level of BLI) and source code (of 1 release for example).
- *Estimation planning level:* Sprint.
- *Estimation purpose:* Project planning.
- *Estimation method:* Planning Poker.
- *Change:* Textual change request (eg, bug report).

2. **Set goals:** To improve estimation accuracy by 10% and to have more transparency in the effort estimation process.
3. **Select impact analysis (IA) and effort estimation (EE) techniques:**

- (a) **IA techniques:** In Figure 4 step 1, the practitioner selects the IA technique described in Gethers et al²⁷ for their context: ie, a textual change request with requirements (user stories or BLI) and code is available. This technique combines informa-

tion retrieval, eg, latent semantics indexing (LSI), with dynamic analysis and mining software repositories (MSR) to determine the impact set of a given change request. This technique can work effectively even when a textual change request is available at the minimum. Additionally, it considers 2 other forms of expert knowledge; a relevant source code entry verified by expert and an execution trace of change request. If any of these additional information is available, the impact analysis is performed more effectively returning more precise impact set.

- (b) **EE techniques:** In Figure 4 step 3, the practitioner selects Planning Poker independent of the IA technique.

4. **Execute:** The execution steps are as follows:

- (a) **Apply and integrate IA and EE techniques:** *IA techniques:* In Figure 4 step 2, the practitioner applies the technique described in Gethers et al²⁷ on the source code, and it returns the impact set at the granularity of the impacted methods in the source code. The practitioner assesses the size as well as the complexity of the impacted parts as medium on the 3-point rating scale (high, medium, and low).

EE techniques: In Figure 4 step 4, the practitioner takes the impact (size and complexity) as input, and uses it to make estimates while playing Planning Poker. During planning Poker, they estimate the effort for the impact as well as the effort required to implement the given change request. The practitioners, for example, mutually agree on an 8-story points estimate for the change request itself and 16 story points for its impact, resulting in 24 story points altogether. This information is stored in an experience database.

- (b) **Analyze quality of estimates:** The quality of the estimates is analyzed by the practitioners during the retrospective meetings. The stored information is retrieved from the database, and the estimated effort is compared with the original to find any deviations. For example, the original effort is found to be 32 story points, with the estimated effort being 24 story points. Further investigation of this underestimation reveals that the effort estimated in the study by Nguyen-Cong and Tran-Cao 6.2 did not account for the effort required to test the change request. This information is also stored in the experience base.
- (c) **Adapt process (with feedback):** To minimize the estimation error found in Nguyen-Cong and Tran-Cao 6.2 the practitioners decide to include additional effort to execute related test cases while making estimates for future change requests. This information regarding the estimation error and measures suggested to resolve it are also stored in the experience database to be reused later on in future sprints.

5. **Analyze quality of selection techniques:** At the end of a project release, the selected IA technique, ie, combination of IR, MSR, and dynamic analysis and the EE technique ie, Planning Poker, are analyzed in terms of their suitability and effectiveness for the estimation process. The results and the adapted process suggests, acceptable results: ie, estimation accuracy has improved over

sprints to an acceptable level, and underestimation is found to have improved from 20% to 15%. The practitioners are satisfied with this improvement. They are also willing to perform impact analysis as it is worth investing the additional effort.

6. **Adapt selection of techniques:** As the goals of improved estimation accuracy and increased transparency in the estimation process are achieved, the practitioners decide to reuse these selected techniques for the future releases.

7 | CONCLUSIONS AND FUTURE WORK

First, this paper described the results of an industrial case study done to understand the process of effort estimation in the agile development context. The major results include factors such as the developer's knowledge, experience, and the complexity and impact of changes on the underlying system, which affect the magnitude as well as the accuracy of estimation and are summarized in Section 5.4. Second, on the basis of the findings of the case study, a framework, inspired by the QIP²⁶ was proposed in Section 6.1 with an application scenario described in Section 6.2. This framework is expected to enable continuous improvement towards effective effort estimation in practice through the reuse of knowledge and experience gathered in each iteration. With this framework, the estimation process will become more transparent and the practitioners will be able to make more informed decisions.

Regarding future work, we need a set of comprehensive guidelines to integrate the IA and EE techniques. We also require other application scenarios in different contexts to assess the applicability of our framework. We plan to evaluate, for example, the usability and ease of use of our framework against different application scenarios in an industrial setting.

Furthermore, this framework should not be considered as a decision-making system but rather as a decision-support system. It will facilitate intensive discussion sessions in estimation meetings, leading to more reliable estimates. Teams will be able to manage their sprints and releases better in terms of identifying and analyzing hidden risks in addition to developing a common understanding.

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